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Exploring Perceptions of Blockchain in Cross-Border Workforce Mobility

Gurzhii Anastasiia, Haque Bahalul, Naqvi Bilal, Vuolasto Jaakko, Parkkila Janne, Islam Najmul

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Designing a Blockchain-Based Credential Verification System for Improving Global Mobility of Workforce

Abstract. *While most research focuses on the technical side of blockchain-based solutions, empirical studies with HR professionals and the workforce on verification processes remains limited. This paper aims to resolve these gaps, exploring the issues and processes that need to be considered before utilizing blockchain solutions in recruiting. We collected data using 13 semi-structured interviews among experts who are involved in the applicants' credentials verification process and 1 workshop among applicants. We analyzed the data from interviews using the content analysis approach focusing on 3 themes, namely key motivators, challenges faced and proposed solutions. These were then validated and expanded during a workshop. Based on the collected data we provide essential design requirements for future systems and align them to all stakeholders involved in the process (individuals, certificate issuers and HR department). Our study extends prior research on blockchain-based verification systems, enhancing collaboration between recruiters and applicants.*

Keywords: blockchain, credential verification, recruitment, design requirements.

1 Introduction

Digitalization is becoming increasingly important across various domains due to many reasons such as focus on sustainability in organizations [1, 2], disruption of the competitive landscape [3,4], changes in consumer behavior and expectations [5], structural changes in the business environment [6], etc. Even though the potential of information technologies in this process is vivid, certificates have stubbornly resisted integration into digital platforms. One of the reasons is they exist in different formats in disparate databases and often require labor-intensive manual verification [7]. Additionally, those certificates are still predominantly distributed in physical formats [8] rather than electronically. This situation creates several challenges when verifying the certificates such as reliance on multiple accrediting bodies, time-consuming procedures and administrative processes supplemented with the potential loss or document damage [9].

Fake credentials are seen as a widespread phenomenon. Ezell and Bear [10] discussed that such problems are growing exponentially since producing academic diplomas and certificates from deceptive websites is not a daunting task these days. A recent Forbes [11] research survey revealed that more than 70% of applicants provide false information in their resumes, such as education credentials falsification, previous work experience exaggeration, employment dates and workplaces. However, the fraud is not limited only to educational credentials, but also trust in the whole traditional certification system experiences a significant decline in value and reputation [12]. In addition to compromising ethical position, holders of fake diplomas do not possess the necessary knowledge, which may pose a serious threat in the professional environ-

ment. Recent research highlights the degree of fraud that has permeated in critical sectors such as healthcare [13], aviation [14] and even governmental positions [15].

Blockchain has become a transformative innovation, marking a significant milestone following the advent of the Internet of Things, Artificial Intelligence (AI) and cloud computing [16]. Using distributed data storage, peer-to-peer transmission, cryptographic algorithms and consensus mechanisms, blockchain provides a range of benefits including fraud prevention, data transparency, decentralization, traceability, immutability and trust establishment [17]. In credentials management, blockchain technology has the potential to accelerate document processing, improve talent acquisition and increase productivity during the verification process that leads to more sustainable practices [18]. Some of the research has already focused on blockchain-based solutions to eliminate fraud issues among certificates and diplomas [7, 12, 19]. Despite the potential and existing efforts, the use of blockchain-based verification process during recruitment and admission services is largely unexplored in the IS literature. In particular, there is a lack of empirical evidence for understanding what challenges organizations face while recruiting and verifying educational or work-related certificates. Furthermore, prior IS research lacks identifying design requirements for such verification systems. For example, most studies focus on value-added services and system architecture [20], challenges associated with educational credentials and how digitalization can resolve those issues [21], how blockchain can revolutionize recruiting processes [22] and blockchain in education [23]. Therefore, our objective is to develop a blockchain-based verification system and in this study, we aim to (i) validate and expand the existing challenges that job seekers and employers face during recruitment regarding credential verification through an empirical investigation and (ii) identify design requirements for the blockchain-based credential verification platform development based on the comments and recommendations from potential users.

This research makes four valuable contributions. First, it identifies the main motivators influencing the adoption and usage of blockchain-based credential verification. Second, it identifies the main challenges associated with existing verification processes across various domains and our research provides a set of design requirements that support future development. Third, our findings may attract more attention not only from the business organization perspective but also encourage individuals to initiate the verification and support of emerging technologies from relevant institutions. Fourth, the goal is to navigate the creation of an effective, fair and adaptable hiring process that can meet the organisation's immediate needs and ensure the workforce's long-term sustainability by utilizing emerging technologies.

The remaining part of the paper is structured as follows. In Section 2 we provide a literature review regarding blockchain in the recruiting and verification domain. Section 3 contains data about the method used in this research. Section 4 highlights the results of the data collection and blockchain-based systems design requirement. Section 5 summarizes the whole research alongside limitations and future research direction.

2 Literature background

In recent years, the interest in blockchain has become incredibly high due to its inherent characteristics such as integrity, non-repudiation, transparency, and privacy [24]. In simple words, blockchain is a cryptographically secure protocol that facilitates the creation of an immutable digital data structure where asset transactions are securely shared between parties in public or private peer-to-peer networks. The main attributes of the technology include trust, transparency, and elimination of all potential third parties that are common to traditional systems [25]. While the predominant application of blockchain is associated with financial and supply chain domains, the deployment of the technology in certificate verification and recruitment gained more attention both among businesses and in academia [26].

While existing research reveals attempts and solutions on how to prevent the misuse of official documents, they continue to occur in higher education institutions [23], various cases of document forgery continue to be reported, especially of diplomas, the most important identity document for jobseekers in both the public and private sectors. Even though blockchain offers opportunities for document verification in higher education, it has not been adopted widely yet due to obstacles (e.g. operational issues, cost of new technology, human resource challenges, regulative environment, etc.) [27]. Research is focused on the platforms that generate, verify, and revoke diplomas [28, 29], receive degree verification [30], models on how to design the system and frameworks [31, 32, 33, 34], blockchain-based projects overview in this domain [7, 35] and development of private solutions [36]. The adoption of blockchain technology in higher education institutions provides transparent and online access to public data. The legitimacy of records remains uncompromised and concerns about data integrity are lowered due to the immutability inherent in blockchain. Any ambiguities associated with degree records can be easily resolved and verified on the blockchain system [27].

In today's competitive environment, companies must align their operations with innovative technologies and reshape their existing business models to survive in an ever-changing world [37]. Despite advances in research on hiring, performance management and training, Blockchain's adoption in these areas remains limited [22]. The lack of understanding of Blockchain's impact on human resource management (HRM) has encouraged scholars to further explore how the technology could support hiring practices, performance management systems, and training protocols. Hence, in the literature researchers pay attention to fraud prevention through the interviews and identification of qualified employees [38, 39, 40, 41], certifications, training and verification [42, 43, 44] and overall innovation in the employment relationships [38, 45, 46]. Overall, blockchain technology offers new prospects for improving security, efficiency, cost-effectiveness, and overall quality of HR (Human resources) procedures. Scholars state that the adoption of blockchain-based verification systems is expected to be gradual [37]. In addition, the role expectations of recruiters may also change, with an increasing focus on understanding technology mechanisms, assessing the cultural fit of candidates, rather than just performing routine tasks. Overall, the literature shows that many educational institutions are already using blockchain technology and how it can improve the efficiency, reliability, and decentralization of certificate verification processes.

3 Method

The data was collected in two stages. The first one consisted of semi-structured interviews and second was a jobseeker workshop. In the first stage we carried out 13 interviews representing university admission services, corporate HR managers, and SME representatives. The interviews focused on discussing the challenges and provided us with a detailed overview of the validation process of received documents. In the second stage, we arranged one workshop with jobseekers to complement the findings from the interviews. The workshop data shed light on the drivers, challenges, and requirements for certificate verification from the jobseeker's perspective.

3.1 Interviews protocol

We conducted 13 semi-structured interviews in 2023. The participants of the interview were employees of companies from various sectors and sizes, with different positions and roles in the applicant certificates authenticity validation process. The interviews duration was from 20 minutes to 1 hour. Each interview process was divided into 4 themes: (i) background and problem description in the domain of employment credentials verification, (ii) verification procedure in more detail, (iii) requirements for the new system, and (iv) financial aspects of the platform and potential costs. Overall, the interviews focused on understanding the possible adoption and use of verification services from third-party providers and the common challenges related to the certificate's validation in their domains. The experts' experience was more than 2 years on their current position.

3.2 Co-creation workshop protocol

To identify the challenges faced with current verification schemes, and the design requirements for a prospective system, a co-creation workshop was conducted (Jones 2018). The workshop was attended by job seekers from 10 countries and lasted 90 minutes. During the workshop, the participants were divided into 3 groups. It was intended that each group contained members from different countries for a more detailed discussion of their own certificate verification experiences. The workshop had the following objectives: 1. To identify challenges faced by you in the verification of documents issued from abroad. 2. To develop a concept of an ICT-based solution that helps overcome the challenges.

In line with these objectives, each group was tasked to first identify the challenges they have faced in their respective countries and while considering these challenges develop a concept of how an ICT-based solution could help overcome those challenges. For execution purposes, the following 5-step protocol was followed [47]:

Step 1 - Introducing the workshop goals and objectives: At the beginning of the workshop session, participants were introduced to the key objectives and goals. The workshop facilitators communicated these objectives framed in a "What? How? Why?" outline, using short and simplified statements.

Step 2 – The challenges: This step is intended to explore the challenges that job seekers face in general. This step involved the following activities:

- Identifying the role of document verification in employment.
- Discussing current verification schemes that are used.
- Identify the challenges with the current schemes.

Step 3 – The solutions: During this step, the participants were asked to discuss a prospective solution that will help overcome the challenges identified earlier. The participants were asked to consider the output of Step 2 to devise their version of the solution. While proposing the solution the participants were asked to think about the role of blockchain there. Additionally, the participants were also tasked to document at least one merit and one limitation of their solution.

Step 4 – Voting the best one: Each group presented the solutions formulated in Step 3 and after discussion, the participants were tasked to vote for the one, while considering the challenges identified in Step 2.

Step 5 – Additional requirements and design principles: For the best-voted solutions, the requirements were documented primarily by involving the group that proposed, and discussion among all groups helped generate additional requirements.

3.3 Data analysis and validation

The collected data from both interviews and the workshop was analyzed using content analysis using theorization and annotations of concepts [48] and pattern-matching logic [49]. The data through interviews among experts that are involved in credentials verification was collected, transcribed, and broken down into smaller text units and coded according to identified themes namely key motivators, challenges faced and proposed solutions. The line-by-line coding of the interviews aimed to identify recurring patterns and key concepts common among experts in relation to credentials verification. In some cases, dual coding occurred, which was resolved through discussion and regular meetings between the authors of the current research to reach an agreement and ensure inter-coder reliability. Validation of the interviews data involved the workshop organization to collect additional thoughts from applicants utilizing the same themes. Additionally, we used pattern matching to compare findings from the literature review, interviews, and the workshop. This approach expanded the findings of the study by identifying common patterns and themes. As a result, it contributed to comprehensive answers to the research questions.

4 Findings

In this section, we analyzed the combined empirical evidence from interviews with experts and workshop discussions to answer research queries and fill research gaps. We explore the key challenges associated with the current certificate verification procedure, the contextual elements influencing the motivators that may lead to the new system consideration and propose design requirements out of the collected data to support the blockchain-based verification system development.

4.1 Analysis of interviews data

To analyze the interviews data, we divided the 3 Groups as well namely Group A – recruiting specialists, Group B – representatives from small and medium enterprises that are involved in the hiring process to some extent and Group C – representatives from the admission services from universities. The diversity of viewpoints expressed in the interviews highlights the lack of guidance from the business community that companies developing blockchain-based verification solutions should aspire and conduct rigorous research that examines this critical issue in various domains. Table 2 presents the outcomes of the experts' opinions.

Table1. Analysis of interviews data

Theme	Group A	Group B	Group C
Key motivators	1. Adapting certificates importance for various domains. 2. Faster and simplified search of qualified specialists. 3. Effective verification system 4.Streamlining personal data management.	1. Increased time of skills verification and hiring process. 2. Need to expand the global talent pool and quick verification of qualifications. 3. Applicant-driven verification process and emphasis on practical skills.	1. Need for authenticity and accuracy in the document verification process. 2. Importance of documents' origin transparency and need for data integrity maintenance and improved efficiency.
Challenges faced	1. Personal data fragmentation and difficulty in engaging companies. 2. Varying the importance of certificates and focus on personal experience and skills. 3. Misconceptions regarding blockchain-based solutions. 4. Potential data granularity and complexity in the candidate evaluation just based on the certificates.	1. Time-consuming government procedures for applicants from abroad. 2. Low relevance of educational certificates. 3. Difficulty of companies' motivation to join new systems. 4. Focus on specific domains with sensitive data. 5. Lack of global temper-proof pool of qualified specialists.	1. Time-consuming process of manual document handling. 2. Difficulty in verifying double-degree applicants. 3. Inability to meet increasing application demand and lack of external funding for third parties in the verification process. 4. Complexity in the detection of fraudulent activities.
Proposed solutions	1. Focus more on companies' requirements regarding certificates validation. 2. Offer options for individuals to exclude certain data elements of their experience.	1. Advocate for simplifying visa grant procedures. 2. Assess practical skills and relevant work experience rather than relying solely on diplomas and certificates.	1. Introduce digital tools for document analysis and validation to enhance accuracy and minimize human error. 2. Implement advanced fraud detection tools to identify and prevent fraudulent document submissions.

Table 1. Continued.

	3. Clearly define the tool's purpose as verification rather than selection to manage expectations effectively. 4. Provide a connection with existing HR tools. 5. Add additional services: video ref. or AI-based recommendations to filter the data. 6. Clarify who validates entities and bears the cost, considering various business models.	3. Simplify the verification process; create a solution that requires one-time verification. 4. Offer incentives for companies that participate in the verification process (e.g., access to the database of verified specialists). 5. Empower individuals to initiate the verification process from relevant institutions.	3. Consider various partnerships with verification tool providers to replace costly funding models. 4. Establish clear protocols for verifying documents applicants and consider leveraging partner university verifications.

Hence, experts from Group A provided a detailed overview of the existing hiring process in the EU. The findings highlight the need to reconsider the importance of certificates in various companies and focus more on the practical experience and personality of the candidate. At the same time, an emphasis on the existence of fraud and data fragmentation in the certificates industry highlights the potential of blockchain-based solutions. However, there are still some misconceptions regarding blockchain as a technology among experts. To address this issue, there is a need to clearly position the solution as a tool for the verification, not for the selection of applicants. This is critical for managing the expectations of potential users. Additional integration with existing popular platforms (e.g., LinkedIn) and the introduction of additional services (e.g. video or AI-based recommendations) can improve the efficiency of the verification process and increase the number of users. Overall, the experts' opinions highlight the importance of balancing technological solutions with practical considerations.

Group B highlighted that certificates are not the most important part of the recruitment process which leads to the urgent need for the improvement of skills accessibility, especially in the context of expanding the global labor pool access for companies. The high emphasis on practical skills and the challenges associated with lengthy government verification procedures for foreign workforce supplement the need for the new way of credentials validation. However, companies are still reluctant to adopt new verification systems and the trust in such methods is low. Among possible solutions experts mentioned possible incentives as a reward for companies to join the system and letting individuals initiate the verification process themselves.

Finally, Group C experts are representatives from universities and the main issues that were discussed relate to the applicant for a degree from all over the world and their documents for admission. The requirement to validate all certificates, especially for applicants from abroad, is the complete responsibility of the certificate owner. To improve verification efficiency and eliminate fraud universities suggest providers that students may choose and follow the guidelines of the particular country. Overall, the findings highlight the critical importance of ensuring accuracy and authenticity in the document verification process, as well as transparency as to the origin of documents. Data integrity and efficiency of the admission process improvements are crucial, espe-

cially in the case of manual data processing. We found that universities are open to collaborations with blockchain-based solutions providers and are positive about blockchain technology.

4.2 Analysis of the workshop data

The workshop participants were divided into 3 groups and Table 3 shows the results from the analysis of the workshop evidence and the responses provided by each of the 3 groups during the discussion. Determining the key challenges during the certificates verification process and finding an optimal solution based on the empirical evidence is a dominant concern of this research. The groups presented different perspectives regarding the future system, while the challenges associated with certificates authenticity are common in different parts of the world.

Group 1 focused on the foundational principles of blockchain-based solutions with a strong emphasis on the encryption, transparency, and decentralized nature of the technology. Comments from participants revealed that the described problems of traditional document verification processes, including the need to find authorized agencies, undergo complex apostille procedures and difficulties in transferring documents, highlight the urgent need for innovative solutions and improved verification mechanisms. The proposal of a pilot test run to validate the effectiveness of the blockchain-based solution represents a proactive desire of individuals to participate in the implementation process. This way, we can make sure that any improvements in verification technologies are thoroughly tested and improved before widespread adoption.

At the same time, Group 2 emphasized the importance of collaboration with various certificates providers and authorities to ensure continuous access to the document verification process. Among the solutions they provided were the establishment of a common website and the incorporation of API access. The results of the group discussion revealed the need for the documents verification process and data storage modernization utilizing digital technologies. A huge emphasis should be placed on the participation of multiple stakeholders, including universities and other certificate issuers, ministries of education and foreign affairs to support the solution's collaborative nature. Moreover, potential users and more emphasis on the blockchain as a solution training is key to ensuring smooth use of the new online system alongside a unified website and engaging government agencies.

Finally, participants from Group 3 supported the findings from the previous 2 groups and highlighted common issues. Nevertheless, they extended the challenges part underscoring time-consuming procedures, high costs, and the inconvenience of multiple certification locations and working hours limitations. Moreover, integrating the verification system into governmental frameworks alongside QR codes for quick access can significantly reduce both the time and costs associated with the process for all parties involved.

Overall, all participants showed a profound awareness regarding the challenges related to the traditional and commonly recognized verification processes.

Table 2. Analysis of workshop data

Theme	Group A	Group B	Group C
Key motivators	1. Simplified verification process using blockchain and encryption technology.	1. Digital transformation for document verification. 2. Transparent and simple process. 3. Use of blockchain and encryption technology.	1. Online verification by means of blockchain and other digital technologies. 2. Global acceptance and unified standards for verification.
Challenges faced	1. Finding an authorized agency and sending original documents. 2. Apostille sticker processing. 3. Document transferring challenges and risk of losing documents.	1. Attestation from the university, educational ministry, and foreign ministry. 2. Verification for birth and marriage certificates for the local embassies.	1. Time-consuming verification process. 2. Excessive costs for document processing. 3. Need to visit various places for certification. 4. Restricted schedule (Monday to Friday).
Proposed solutions	1. Develop a system that will enable a criteria-based verification. 2. Use digital copies only, no printed documents. 3. Implement API access for verification and authentication. 4. Integrate payment mechanisms through apps and local currencies systems. 5.Conduct a pilot program to test the effectiveness of the proposed solutions.	1. Establish a common website for document uploads. 2. Include government entities in the system. 3. Collaborate with various service providers like embassies and universities. 4. Ensure online services and availability of digital documents 24/7. 5. Implement API access for verification. 6. Focus on user training to ensure a smooth transition to system.	1. Develop an online application system available worldwide. 2. Implement the use of QR-codes. 3. Streamline the certification process to reduce time and costs by the system integration into the governmental systems. 4. Enable online verification without the physical presence. 5. Suggest conducting a feasibility study to assess the practicality.

The proposal to conduct an in-depth case study demonstrates a willingness to thoroughly evaluate the practicality and effectiveness of blockchain-based solutions before implementation.

4.3 Proposed design requirements for a blockchain-based system

The diversity of viewpoints heard at the workshop emphasised the lack of guidance on the level of quality and convenience of the traditional certificate verification process. The workshop data also revealed the need for rigorous research examining this critical issue in different contexts in the recruitment process. We have identified that the challenges of certificate verification are related to lack of centralized storage and verification, transparency in the whole process, lack of flexible and user-friendly mechanism for verification, time consuming nature, lack of coordination among the verification entities and lack of proper data protection mechanism. Upon investigation we have identified several design requirements that can contribute to the design of a

blockchain-based system. These requirements can act as guidelines for the system development, even though new requirements will likely arise during the development of such a system. Therefore, there is need for further testing and validation in the future from practitioner's perspective. The requirements that we propose are as follows:

DR1: **Inclusive solution**. The proposed solution should include all relevant parties involved in the verification process, so that, the user can use this service as one stop solution for any of verification from all required and relevant entities.

DR2: **Fast verification.** The proposed solution for a Blockchain-based credential verification system should provide a quick and flexible verification mechanism so that the users will be able to reduce the time needed for verification in the current process.

DR3: **Unified interface.** The proposed system should provide the users with a single system/interface that can be used for uploading, verifying and downloading relevant documents so that, users will not have to rely on/go through multiple parties again.

DR4: **Flexible data sharing.** The proposed system needs to have a flexible data sharing mechanism such as QR code, so that, users can share this QR code with relevant parties for necessary verification.

DR5: **Payments' flexibility**. The system needs to incorporate flexible payment mechanisms such as integration with the local currency and credit card for international payment so that users can use this system from any place they require with any flexible payment method to choose from.

DR6: **Data protection and privacy**. The proposed system should include proper data protection, privacy mechanisms to comply with existing laws on data privacy (e.g. EU's GDPR) so that, users' valuable data is not hampered in case of system failure.

DR7: **Interoperability**. The proposed system should include interoperability among the credential providers, so that, the users can easily upload and download their document from any platforms.

DR8: **Scalability**. The proposed systems need to be scalable so that, it can efficiently store and analyse increased number of users, especially if more companies are willing to participate in the system.

DR9: **Transparency and integrity**. The proposed system needs to facilitate the companies about the transparency and integrity of the credential provider and their authenticity.

DR10: **Bonus/reward model.** The system can introduce incentives for participating and using this service, therefore more users (both companies and individuals) are willing to participate in the ecosystem.

DR11: **Individuals' empowerment.** The system developers should empower individuals to initiate verification themselves and provide options to connect with certificate issuers in a quick and simple manner.

4.4 Synthesized framework of requirements

Based on the insights obtained from the empirical data analysis, we identified 3 main parties involved in the verification process in the recruiting context: individuals

(certificate holders), certificate issuers, and HR departments/admission services office (Figure 1). We provide a simplified framework that addresses important requirements for every stakeholder involved in the process. In this case, individual entities that initiate the verification from relevant institutions, certificate issuers confirm the authenticity of the document, and HR specialists validate the outcomes of the verification process. The challenges section explains the main constraints that every party involved faces; design requirements are meant to resolve those issues and make the system more attractive to potential users. Our findings complement the existing progress in this domain [7, 12] by focusing on the empirical data and the importance of such system in the verification domain.

Overall, we found that individuals focus on convenience, price and efficiency, emphasizing the need for a comprehensive, fast and secure verification process of their documents. Certificate issuers emphasize functionality and scalability, with a huge focus on interoperability, simplified processes and efficient data management. HR departments and admission service representatives prioritize efficiency and reliability, looking for rational processes, fast validation and transparent, reliable validation results that will be accepted by other parties without an additional timely verification request. Understanding these priorities is critical to developing a solution that meets the needs of all stakeholders and maximizes the benefits of blockchain technology for certificate verification.

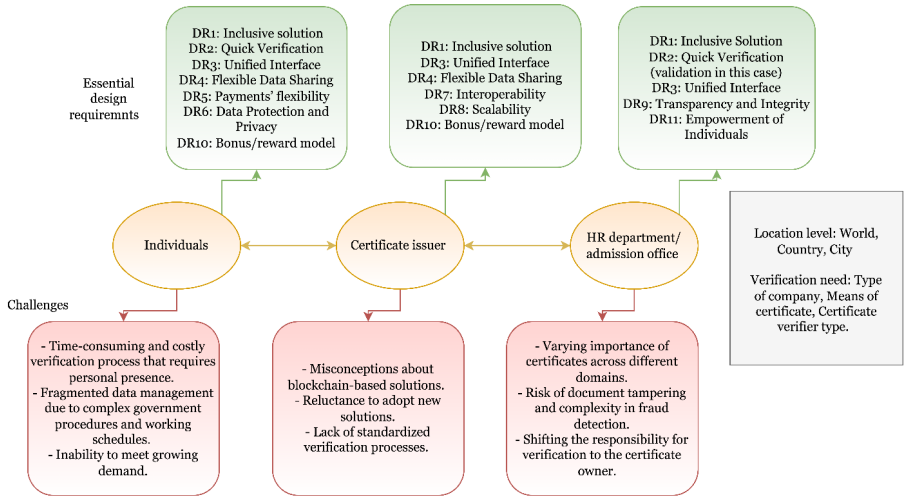


Fig. 1. A framework for scoping blockchain-based verification solutions

Additionally, we found that the general adoption of blockchain-based verification systems depends on several factors. First, individuals who initiate the verification from relevant institutions and encourage other parties to consider alternative options. Second, certificate issuers may collaborate with certificates verification providers to eliminate time-consuming procedures for their students in the future. Finally, HR or admission departments could allow individuals to be more flexible in case of choosing verification methods. Fourth, verification systems providers should unite all par-

ties involved by providing bonus systems, a pool of talented and verified specialists and the creation of viable business models without a stick focus on one stakeholder only.

5 Implications

The paper has several contributions. First, even though numerous papers discuss the role of blockchain in certificate verification and recruiting in particular [7, 12, 19, 23, 27], the overall knowledge regarding the blockchain-based system in the certificate verification process is still limited. While scholars [31, 32, 33, 34, 36] focus on the technical perspective and the development process, we provide a discussion with experts from various domains and job applicants to reveal challenges and design requirements based on their feedback. Second, the findings complement the existing research with empirical data that can be expanded in the future and encourage scholars to focus more on blockchain-based solutions not only in the verification domain but also in the technology adoption in educational institutions. Third, the investigation revealed several challenges for credential verification, especially when a job seeker is applying from a different country and has degrees earned in a different region than that of the employer's country. In this case, job seeker must go through multiple steps and visit multiple verifier entities. Issues have been noticed while recruiting people from around the world in terms of fake credentials, identifying the verifier entities, and establishing the integrity and trustworthiness of the credential provider. Moreover, it is also time-consuming for the employer since they must go the extra mile for verification. Fourth, we have outlined and identified the novel research challenges from both job seekers' and employers' perspective. The data also allowed us to propose design requirements for developing a trustworthy, transparent, and efficient credential verification system.

6 Conclusion

In this study we identified the existing challenges that job seekers and employers face during recruitment regarding credential verification and identified design requirements for the blockchain-based credential verification platform development based on the comments and recommendations from potential users. The current research derived the following conclusions. First, we empirically validated the importance of a blockchain-based verification system among certificates' owners and career experts. Second, based on the data collected through interviews and the workshop we identified essential design requirements that will support future systems development. Third, the deeper insights into HR managers' and business owners' perspective in the case of verification revealed a huge load on applicants. This leads to the significance of focusing not solely on business owners and HR departments but also towards involving individuals and certificate issuers in the adoption of blockchain-based verification systems.

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